

Comparing the role of delocalized states in quantum annealing and exciton diffusion in light-harvesting systems.

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Abstract

This thesis presents a comparative analysis of the role of delocalized quantum states within two frameworks: quantum annealing and exciton diffusion in light-harvesting systems.

The term “delocalized state” refers to a state in which a quantum system (such as an electron) exists in superposition, i.e. in a spread-out probability distribution over multiple positions or energy levels simultaneously.

On the one hand, light-harvesting systems, particularly those found in nature as in photosynthesis utilize exciton diffusion for efficient energy transfer. This process involves delocalization, wherein excitons exist in a superposition state, and explore multiple pathways toward reaction centers for photosynthesis concurrently, thereby optimizing energy transfer.

On the other hand, quantum annealing is a computational process that harnesses quantum mechanics to solve complex optimization problems. During the anneal, the system at hand transitions from an initial ground state (usually a simple one) to a final state that encodes the solution to the optimization problem. This process relies in part on the principle of superposition, which allows the system to exist in a delocalized state, and hence explore multiple pathways through an energy landscape simultaneously—potentially leading to faster convergence toward the optimal solution.

Despite their different applications, the role of delocalized states in both systems is strikingly similar. This project will delve into these parallels, aiming to uncover new insights that could inform both the study of energy and information transport in quantum technologies. Through this research, we hope to contribute to the broader scientific understanding of delocalized states and their role in physical and biological systems as well as shed light on the parallel evolution of scientific knowledge in different but interrelated domains. Discussions of the physics of delocalized states in both cases are followed by derived implications for science and technology policy.

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1 Introduction

1.1 Motivation - quantum computing and quantum energy science

In recent years, the quantum technology (QT) landscape has witnessed unprecedented growth, with substantial investments—\$2.35 billion in 2022—flowing into quantum startups [1]. While quantum computing (QC) has dominated this surge—funneling 75% of these investments [1]—this thesis aims to redirect attention to other facets of quantum technologies, particularly those relevant to energy harvesting applications. Our motivation lies in drawing parallels between energy transport and information propagation within quantum systems as, after all:

“Information is physical.”

- Rolf Landauer

In quantum computation, a two-level system (qubit) is seen primarily as an information carrier. In light-harvesting systems, a comparable two-level system may be seen primarily as an energy carrier.

In this report, we will look at energy through the lens of light-harvesting systems, and look at information through the lens of quantum annealing. We will show the similarities, and differences, in the treatment of delocalized states in these seemingly different use-cases.

Overview of the report

We will first consider exciton diffusion, where excitons—electron-hole pairs that can be viewed as excitations spread across light harvesting complexes towards a reaction center (RC). We will simulate this propagation through a chain of two-level systems (TLSs)—a simplistic model that may later be extended to higher dimensions. Secondly, we will consider quantum annealing (QA). The motivation for QA is the need to find the global energy minimum within an energy landscape, that is set up in such a way that it represents the optimal solution to a complex optimization problem. We will lay out the formalism for this optimization process, to better understand how it works. Then, we will put it to the test by attempting to solve the “Shortest Path Problem”. Finally, we are interested in comparing the efficiency of both of these transfer processes.

In this thesis, we will break down both mechanisms, paying special attention to the role of delocalized states. Such comparative studies are important as—like some observers have argued (Aleta et al. 2018) [2]—knowledge is becoming increasingly compartmentalized, with synergies and complementarities across fields being overlooked.

1.2 Fundamentals and notation

1.2.1 Two-level systems

Two level systems (TLSs) are one of the simplest quantum systems (considering the one-level system to be trivial). They serve an important function as many physical systems may be reduced to TLSs, making them conceptually and computationally more tractable. A TLS, as its name indicates, has two possible states. In quantum computing, we call a **qubit** or quantum bit a basic unit of quantum information i.e. the quantum analog of a classical bit—a binary digit [3]. A qubit may be represented by a TLS. In a classical system, a bit takes one of 2 possible

values, 0 and 1, at a time, whereas a qubit is allowed to be in a linear superposition of the two basis states of the 2D Hilbert space in which it lives. In general, the state of a qubit is expressed as a linear combination of $|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, the qubit's ground state, and $|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, the qubit's first and only excited state: its two orthonormal basis states—*computational basis states*. We call the state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle \quad (1)$$

a superposition state.

Throughout this report we will refer to an **excitation** as the addition of a discrete amount of energy to a TLS. In particular, when dealing with qubits, we will call this excitation the transition from the state $|0\rangle$ to $|1\rangle$. We will also tolerate a somewhat misuse of this term and refer to the energy transfer between neighboring TLSs as the *hopping* of an excitation from one system to the next.

Brief aside on measurement

When we measure a qubit in the state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, we get either the result 0, with probability $|\alpha|^2$, or the result 1, with probability $|\beta|^2$. The law of total probability thus implies a normalization condition on these factors. In this case $|\alpha|^2 + |\beta|^2 = 1$.

Suppose now that we have two TLSs. This system now has 4 computational basis states $|00\rangle, |11\rangle, |01\rangle$ and $|10\rangle$. Hence, a couple of qubits can exist in a superposition of these states, and the state vector representing these qubits is of the form

$$|\psi\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle \quad (2)$$

Similar to the case for a single qubit, the measurement result x ($= 00, 01, 10$ or 11) occurs with probability $|\alpha_x|^2$, with the state of the qubits after the measurement being $|x\rangle$.

These two qubits are thus represented by a superposition state vector in $2^2 = 4$ dimensional Hilbert space. In general, n qubits are represented by a superposition state vector in 2^n dimensional Hilbert space.

“Hilbert space is a big place!”

- Carlton Caves

1.2.2 Delocalized states & Dicke states

In the classical picture, an electron may move along a lattice as a point-like object, defined by its position x and momentum p . In the quantum mechanical picture, the state of this electron is defined by a wave-function $\psi(x, t)$, where x is position and t is time, whose square modulus $|\psi(x, t)|^2 = \psi^*(x, t)\psi(x, t)$ is a probability density of the electron having a given position at a given time. According to the uncertainty principle, it is not possible to have simultaneous exact knowledge of the position and momentum of our particle, i.e. $\Delta x \Delta p \geq \hbar/4\pi$. In some sense, a particle described by quantum mechanics can never be fully localized with absolute certainty (if we want to know anything about its momentum), as this would violate the uncertainty principle. However, throughout this thesis, we will say that a particle is **localized** when most of its probability density is concentrated within a constant small region of space. The notion of *small*

will depend on the system at hand. In the case of a lattice of TLSs we will consider throughout this thesis, this region is considered to be small if it is smaller than the distance between two TLSs in the lattice.

Let us consider two TLSs, both initially excited. Our system is thus initially in the state $|11\rangle$. Let us investigate the ways in which this state could emit both of its excitations, i.e. the possible pathways to $|00\rangle$.

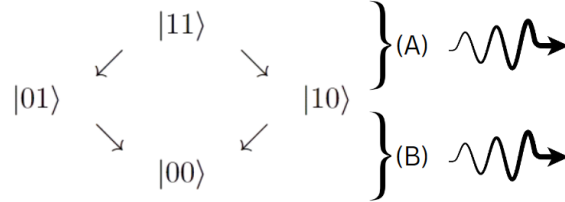


Figure 1: Possible pathways from $|11\rangle$ to $|00\rangle$ through discrete emission steps.

In this case, after the process of (A) seen in figure 1—where only one excitation is in our system—this excitation is not localized at the first or second TLS, but delocalized over the two. So our system, at the second step in the diagram, is in the superposition state $(\alpha|01\rangle + \beta|10\rangle)$, where α and β are normalization constraints such that $|\alpha|^2 + |\beta|^2 = 1$. This state can be called a **delocalized state** [4]. It is helpful to imagine this intermediate step as the excitation simultaneously exploring two possible pathways. In other words, in a delocalized state, the spatial domain is coherently covered by the excited state (see section 1.2.3 for a definition of coherence).

What's more, the states $|00\rangle$, $|11\rangle$, and $(\alpha|01\rangle + \beta|10\rangle)$ in the example above correspond to so-called **Dicke states**. A Dicke state is a kind of delocalized state, in which the delocalization is symmetric. Formally, it may be defined as follows

Definition 1 (Dicke State [5]). *A Dicke state is an equal-weight superposition of all n -qubit states with a specific Hamming Weight k . Where the Hamming weight is the sum of the binary variables in our system, i.e. if $\underline{s}$ is an N -dimensional vector of binary variables (so we have either $s_i = 0$ or $s_i = 1 \ \forall i$)*

$$wt(|\underline{s}\rangle) = \sum_{i=1}^N s_i$$

A Dicke state is then defined as

$$|D(n, k)\rangle = \frac{1}{\sqrt{\binom{n}{k}}} \sum_{wt(|x\rangle)=k} |x\rangle$$

These states induce many interesting properties and, under certain conditions, can lead to collective quantum effects such as Dicke enhancement (sometimes referred to as superradiance, superabsorption, and supertransfer in different contexts [6, 7]) which drive fast dynamics of interest. Essentially, if each TLS in our system is coupled to a shared field mode, this will boost coupling strength while preserving other quantum effects. In general, delocalized states—and the special case of Dicke states—are a defining property of quantum systems. When they come into play, these states, characterized by their spatial spread, have the potential to speed up

diverse processes from energy transport in light-harvesting systems to information transport in quantum annealing.

1.2.3 Coherence & Decoherence

We are interested in defining coherence and decoherence as we will see that they play a role in efficient energy transfer in light-harvesting systems [8] and quantum annealing.

Coherence refers to the ability of a quantum system to maintain a well-defined phase relationship between different states in a superposition. For instance, in the case of the state

$$|\psi\rangle = \omega_a|a\rangle + \omega_b e^{i\phi}|b\rangle \quad (3)$$

if ϕ is well-defined, then there is perfect coherence between $|a\rangle$ and $|b\rangle$. Moreover, as long as there is no interaction between the system and its environment, this coherence should be conserved. Coherence is also sometimes related to the presence of off-diagonal elements of a density matrix that describes a system's time evolution.

We saw that a qubit can exist in a superposition of $|0\rangle$ and $|1\rangle$ states, represented as $\alpha|0\rangle + \beta|1\rangle$ where $|\alpha|^2 + |\beta|^2 = 1$. Coherence ensures that the relative phase between these states is preserved, allowing for constructive and destructive interference in quantum algorithms. Without coherence, the qubit loses its quantum behavior, and quantum computations become meaningless. Hence, coherence is an interesting quantum resource [9], whose measure might serve as a potential proof of *quantum advantage* (the ability for a quantum computer to perform a given task more efficiently than its classical counterpart).

Decoherence is the process by which a quantum system loses its coherence due to interactions with the environment. It is one of the primary challenges in building and operating quantum computers [10]. Decoherence can occur through various mechanisms, such as thermal fluctuations, electromagnetic interference, or imperfections in quantum gates. Preserving coherence requires careful isolation, control, and error mitigation.

Looking ahead

In this introduction, we have looked into key underpinnings of quantum mechanics: defining the two-level system, delocalized states, Dicke states, coherence and decoherence. We are now equipped with the proper tools to investigate quantum effects in exciton diffusion and quantum annealing. Let us then begin by looking into exciton diffusion in light-harvesting systems.

2 Exciton diffusion in light-harvesting systems

In this section, we will explore the role of delocalized states in exciton diffusion. We will first define the terms relevant to this section and explore historical and technological relevance. We will then turn to a more quantitative picture of diffusion, modeling the dynamics of state transfer through a chain of two-level systems (TLS).

2.1 Defining exciton diffusion

Exciton diffusion is a mechanism of energy transfer in light-harvesting systems. Plants absorb photons, and excitons transport energy to where it can be stored chemically for metabolic processes.

Excitons are electrically neutral quasiparticles and can be regarded as elementary excitations at the molecular level that can transport energy without transporting net electric charge. They arise from the interaction between electrons and electron holes. When an electron and a hole are close enough, their combined energy state forms an exciton.

In general, we define diffusion as the *random* motion of particles in space that leads to spreading from the areas of high concentration to the areas of low concentration. In the context of exciton diffusion by light-harvesting systems, diffusion describes the movement of excitons through a network of molecules. Depending on the physical characteristics of the molecular aggregate, excitons can be localized on a single molecule, in which case one speaks of localized excitons (or sometimes simply excitation). Alternatively, excitons can be delocalized coherently over several molecules, effectively composed by a superposition of excited states belonging to different pigments [11, 12, 13]. The transfer of energy then occurs via a strong dipole-dipole coupling between pigment molecules [14]. When molecules are positioned closely to each other, these inter-molecular electronic interactions are strong enough that the quantum mechanical superposition principle causes excitation to be shared coherently, resulting in a redistribution of dipole strength among transitions, and changes in orientation of transition dipoles.[12]

The study of underlying quantum effects in light-harvesting systems is an active field of research, in fields ranging from Biology to Physics, to Electrical Engineering. I invite the interested reader to dive deeper into the history of this study by reading the following insightful review papers [15, 12, 16]. This fascination has even led to the publication of popular science articles [17], some going as far as to claim that:

When it comes to photosynthesis, plants perform quantum computation.

or with headlines such as

Plants ‘seen doing quantum physics’ [18]

While such claims are certainly sensationalist in their wording, they do refer to actual research, as summarized in this 2014 article by Seth Lloyd [19]. The argument in a nutshell is that fast coherent dynamics are responsible for the remarkably high efficiency of photosynthetic processes.

The realization that a biological process, normally understood to operate in a classical, thermodynamic limit, can utilize quantum-mechanical superposition states and interferences is quite baffling. Moreover, this recognition implies that we can harness ‘*quantum weirdness*’ in other nanoscale systems, including possibly engineered materials, outside of pristine laboratory conditions such as extremely low temperatures [12]. The following sections will explore some of these potential applications in greater detail.

2.2 Concrete motivation for studying exciton diffusion

Solar power is abundant on Earth, but man-made light-harvesting systems face many technical and intrinsically physical limitations. The transport process in light-harvesting systems (found in Nature) has been found to reach yields of above 95% [20], meaning that the energy of more than 95% of all absorbed photons ends up being metabolized rather than lost to re-emission. The potential technological implications of a better understanding of photosynthetic organisms’ solar energy transport have motivated the study of these organisms. A key challenge in the design of organic solar cells has been the rate of exciton diffusion: if that rate is too low, then much of the energy that got initially absorbed from photons gets lost to re-emission before it can reach

the cell’s charge separation barrier. A better understanding of these processes might allow us to engineer efficient transfer of excitation energy from sites where photons get absorbed to sites where that energy can be turned into electrons. Potential applications of accelerated exciton diffusion range from organic photocells to solar fuels and other light-harvesting techniques [21, 22, 12, 23, 24].

3 Simulating exciton diffusion: Chain of 2-level systems

In this section, we explore how the *random* motion of exciton transport occurs in light-harvesting systems, focusing on 3 models of excitation transfer through a 1D chain of 2-level systems.

Aside on physical limitations of the model

As is often the case in physical models, reducing the intricacies of a network of chromophore molecules to a 1-dimensional chain of two-level systems (TLS) is a rather crude simplification of the complex structure that is a light-harvesting system. However, it has been suggested that this simplification has some physical relevance, as it has been used in the literature [25], so we will content ourselves with it for now. Such models can be expanded to two and higher dimensions, but key behaviors can already be observed in the 1-dimensional version.

Furthermore, we are working under the assumption that each molecule in our chain is identical. This is not generally the case in light-harvesting harvesting systems, as chromophoric complexes are—as their name suggests—complex assemblies of non-identical chromophores. Such assumptions—and the resulting gaps—need to be kept in mind when comparing simulations results as developed below to dynamics observed in nature.

Let us start by considering a simplified picture, in which two neighboring chromophores C_1 and C_2 are off resonance. For the sake of simplicity, let us imagine that the excitation in our system is C_1 at $t = 0$, and will transfer to C_2 at $t = \tau$. Let us further assume that C_1 (resp. C_2) has energy E_1 (resp. E_2), with $E_1 > E_2$. When our exciton moves from C_1 to C_2 , this energy difference ($E_2 - E_1$) can be released as a phonon. This simplified picture provides insight into the limitations of considering our chromophoric complex as a simple chain of TLS. In fact, the various processes leading to phonon-emission has been studied in depth [26, 27], but its complexities go beyond the scope of this thesis. Hence, for the purpose of understanding state-transfer dynamics, we will content ourselves with this model—all the while acknowledging its limitations.

Motivation for the considered models of exciton diffusion

The models we will explore are:

1. Heisenberg nearest neighbor interacting spins model for state transfer
2. continuous-time quantum walk
3. Classical random walk

Exciton transfer mechanisms can often be described by a semiclassical Förster theory which involves incoherent hopping of the excitations between sites. In 2012, *Feron et al.* showed that exciton hopping in the Förster model can be fully described in terms of random walk [28]. More

generally, the exciton motion inside a molecular network is often described by nearest-neighbor steps by a random walker [29]. Feynman first introduced the concept of a quantum walk when modeling the dynamics of a quantum particle on a lattice [30]. Later, in 1980, J. Klafter and R. Silbey introduced the idea of continuous-time random walk to describe the transport of excitations on a lattice [31]. However, the formalisms of these walks were only later defined. In 2008, *Mohseni et al.* further developed a quantum walk approach, as a natural framework for incorporating quantum dynamical effects in energy transfer, as opposed to this classical random walk picture [32]. They demonstrated that this quantum random walk provided an appropriate mathematical framework for studying energy transfer. This walk was also revealed by imaging by *Akselrod et al.* in 2014 [33].

Hence, to learn more about the role of delocalized states in exciton diffusion, we will first look at a general state transfer model: the Heisenberg model for next-neighbor interactions. We will then replicate a simplified diffusion process more specific to light-harvesting systems by modeling a quantum random walk, along a chain of two-level systems (qubits). We will gain further insight into the matter by comparing a classical random walk to this quantum random walk.

Furthermore, it is common to think of the dynamics of energy transfer in terms of the evolution of populations—probabilities that excitation has reached the various sites in a light-harvesting complex [12]. This will be our point of comparison for the three models.

3.1 Heisenberg model nearest-neighbor interaction

Let us first abstract from the context of light-harvesting systems to investigate quantum state transfer through a 1D chain of TLS. We will first consider our system to be a spin-chain. Ultimately, the precise nature of the two-level systems (TLS) we are considering does not matter, because of the following fact.

Remark 3.1 (Isomorphic 2D Hilbert Spaces). All two-dimensional Hilbert spaces are isomorphic.

Proof. Result of the Riesz-Fischer theorem, see [34] for full proof. $\square$

Thanks to 3.1, we can choose to model the TLS in our chain as any well-studied TLS, such as spin 1/2 particles. The spin chain is a well-studied model [35, 36], and the Heisenberg model is particularly successful in describing 1D spin chains. Furthermore, Heisenberg chains have been shown to act as a quantum wire to transfer a quantum state. This motivated our approach, as we attempt to describe the transfer of an excitation along a chain of TLS [37, 38]. The model is all the more interesting, as we consider these quantum wires as both information and energy carriers.

The Heisenberg magnet is described by a nearest neighbor Hamiltonian of the form:

$$H = \sum_{i=1}^n (J_x \sigma_x^{(i)} \sigma_x^{(i+1)} + J_y \sigma_y^{(i)} \sigma_y^{(i+1)} + J_z \sigma_z^{(i)} \sigma_z^{(i+1)}) \quad (4)$$

where σ_α^i represents Pauli- α operator acting on the i^{th} qubit in our n-qubit chain.

Various *versions* of this model exist, in which the coupling coefficients take different values. Historically, the Heisenberg XXX model has provided a solid background and a strong motivation

for many elegant studies into the algebraic structures and symmetry of the model. In this model, we take $J_x = J_y = J_z = J$, hence the Hamiltonian becomes:

$$H = J \sum_{i=1}^N (\sigma_x^{(i)} \sigma_x^{(i+1)} + \sigma_y^{(i)} \sigma_y^{(i+1)} + \sigma_z^{(i)} \sigma_z^{(i+1)}) \quad (5)$$

Another interesting version is the XXZ model (where $J_x = J_y = J$ and $J_z = 0$) described by a Hamiltonian of the form:

$$H = J \sum_{i=1}^{N-1} (\sigma_x^{(i)} \sigma_x^{(i+1)} + \sigma_y^{(i)} \sigma_y^{(i+1)}) \quad (6)$$

The model best fit to describe our system is the XXZ model (sometimes also referred to as XY model), in part because it is closest to its adjacency graph. This will become clearer in the last part of this section.

Let us first consider a chain of $n = 2$ spins, with $J = 1/2$. We initialize our system to have an excitation in the first qubit in the chain i.e. in the state $|\psi_0\rangle = |1\rangle \otimes |0\rangle$. We simulate the time-evolution of the XXZ Heisenberg Hamiltonian, in python using the quantum dynamics package *QuTiP*. We then plot the time-evolution of the probability amplitude of being at each node in the chain.

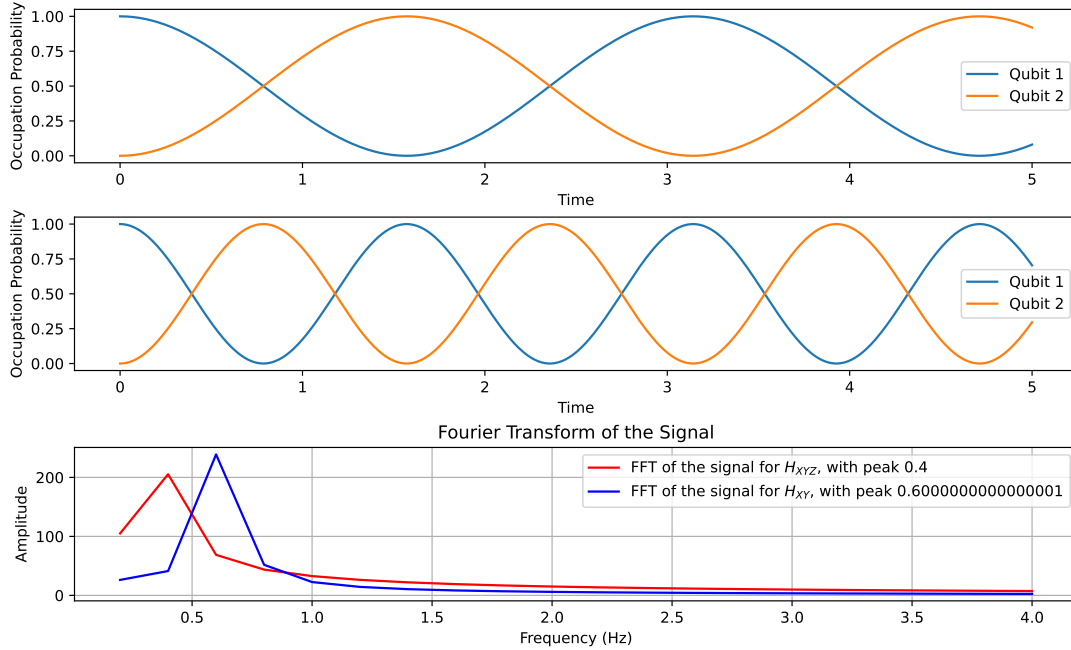


Figure 2: Time evolution of occupation probability in chain of 2 qubits under H_{XXZ} i.e. with TOP: $J_x = J_y = 0.5$, and $J_z = 0$, BOTTOM: $J_x = J_y = 1$, and $J_z = 0$ coupling strengths and transfer frequency

As expected figure 2 shows a periodic transfer of the probability amplitude of the excitation at a given qubit in the chain. Hence, at a given time, the probability of the excitation being at the end of the chain will depend on the time at which we observe our system. This serves

as a kind of sanity check as this is expected behavior, but we are interested in observing the behaviors of more complex, specifically longer, chains. Furthermore, the Fast Fourier Transform (FFT) of the signal tells us that the frequency of transfer will depend linearly on the coupling strength between neighboring qubits, as expected. This highlights the importance of strong dipole-dipole coupling in fast excitation transfer, which in turn is affected by the geometry of the system in question (e.g. distance and orientation of molecules).

Let us now consider a chain of $n = 3$ spins, keeping $J = 1/2$, initially in the state $|\psi_0\rangle = |1\rangle \otimes |0\rangle \otimes |0\rangle$. Again, we simulate the time-evolution of the system according to the Schrödinger equation in Python, and plot the resulting occupation probabilities in figure 3.

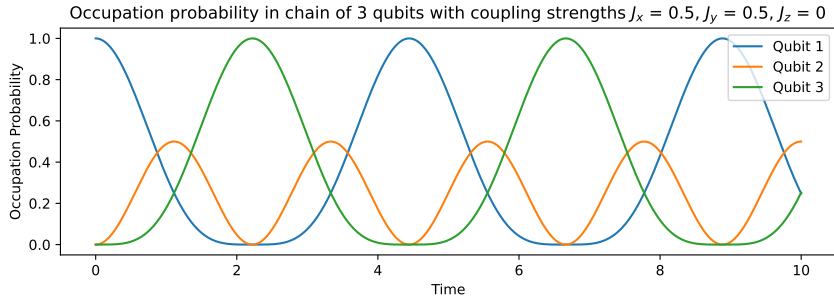


Figure 3: Time evolution of occupation probability in chain of 3 qubits under H_{XXZ} i.e. with $J_x = J_y = 0.5$, and $J_z = 0$ coupling strengths

The dynamics of our system are much more interesting now. The excitation is initially at the first qubit in the chain with probability 1, and after a certain time should be at the 3rd qubit in the chain with probability 1. However, the probability of being at the second qubit in the chain is never greater than $1/2$. This goes against classical intuition and is due to the fact that the states are delocalized. Indeed, the excitation does not ‘hop’ from one qubit to the next, but explores the possible pathways simultaneously. So, there are several possible paths for the excitation to get through the second qubit. What is meant by this is that when the probability of being in the state $|010\rangle$ is maximal (near $1/2$), the probability of being in the states $|100\rangle$ or $|001\rangle$ is non-zero, under the action of the nearest neighbor Hamiltonian.

For longer chains, as is illustrated in figure 4, the dynamics become far more complicated. However, we note that the probability of being at the last qubit in the chain still grows very quickly, and—for chains of length smaller than three—is close to 1. These state transfer dynamics have been studied in depth, and it has been shown that perfect state transfer is only possible for chains of lengths smaller than three [38]. It is interesting to have a closer look at the speed with which the excitation reaches the last qubit. In the case of a light harvesting system, this corresponds to the probability of the excitation initially created by the absorption of a photon to make its way to the reaction center (RC), where the absorbed energy is metabolized. In the context of designing efficient organic solar cells, we want this occupation probability to grow quickly regardless of the length of our chain. Here, fast rates of exciton diffusion enable thick absorption layers, which maximize the percentage of photons captured without losing many of them to re-emission.

The Heisenberg model serves as a general model for state transfer in a chain of TLS, when considering only next-neighbor interaction. Let us now try to gain a deeper understanding of the particular dynamics of exciton diffusion by diving into the quantum walk model.

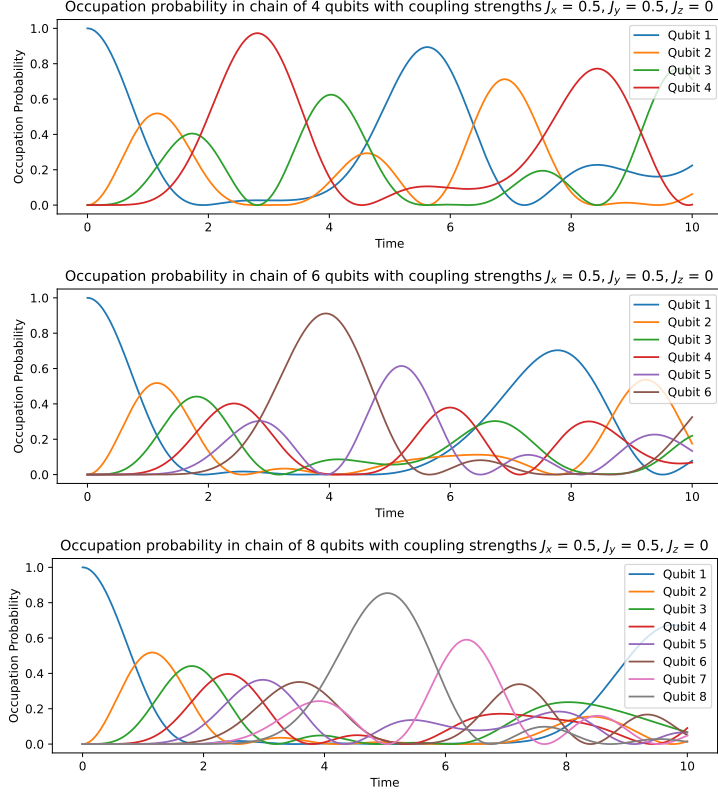


Figure 4: Time evolution of occupation probability in chain of 4, 6, 8 qubits resp. under H_{XXZ} i.e. with $J_x = J_y = 0.5$, and $J_z = 0$ coupling strengths

3.2 Quantum random walks

In a light-harvesting system, a photon is absorbed by a chromophore molecule and an excitation is created. This excitation is then transferred through the network of molecules known as the light-harvesting complex until it reaches the so-called reaction center (RC), where its energy may be trapped to access the cell's metabolism. Now that we have explored general state transfer dynamics in qubit-chains, let's have a closer look at the dynamics of exciton diffusion. The exciton propagation mechanism may be viewed as similar to that of a quantum walker on a graph where each vertex point represents a lattice site, with the propagation route determined by the edge joining one vertex to another. The first step in setting up this model is to understand the mechanism of the quantum random walk.

Discrete quantum random walks

It is useful to first visit the discrete quantum random walk as a preliminary to understanding its continuous-time counterpart. The quantum random walk was introduced by *Aharonov et al.* in 1993 [39]. To understand this model, we will base ourselves on the introduction made by Julia Kempe [40].

Let $\mathcal{H}_p$ be the n -dimensional Hilbert space spanned by the positions of the particle. For a line with grid-length 1 (a chain) this space is spanned by basis states $\{|i\rangle : i \in \mathbb{Z}\}$. We will restrict ourselves to a finite line, so our space is spanned by the basis states $\{|i\rangle : i = 0, \dots, n\}$,

$|i\rangle$ corresponding to a particle localized in position i in the chain. The position Hilbert space $\mathcal{H}_p$ is augmented by a ‘coin’-space $\mathcal{H}_c$ spanned by two basis states $\{|+\rangle, |-\rangle\}$. States of the total system are in the space $\mathcal{H} = \mathcal{H}_c \otimes \mathcal{H}_p$. The conditional translation of the system, or shift operator (taking the role of $\exp(-2iS_z \otimes P_l)$) can be described by the following unitary operation

$$S = |\uparrow\rangle\langle\uparrow| \otimes \sum_i |i+1\rangle\langle i| + |\downarrow\rangle\langle\downarrow| \otimes \sum_i |i-1\rangle\langle i| \quad (7)$$

where the index i runs over the indices of the qubits in our chain.

Remark 3.2. We use the $|\uparrow\rangle, |\downarrow\rangle$ notation rather than the $|0\rangle, |1\rangle$ notation we have been using until now to represent the qubit basis states to avoid confusion with the basis states of $\mathcal{H}_p$.

We will then use, to begin with, the Hadamard coin as our unbalanced unitary coin:

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad (8)$$

The quantum random walk of T steps is then defined as the transformation U^T , where U , acting on $\mathcal{H} = \mathcal{H}_c \otimes \mathcal{H}_p$, is given by

$$U = S \cdot (C \otimes I) \quad (9)$$

For the purpose of this thesis, we will not dive any deeper into this model, but I invite the interested reader to have a look at *Google Quantum AI*’s notebooks on discrete quantum walks.

Continuous-time quantum random walk (CTQW)

The continuous-time quantum walk was first considered by Farhi and Gutmann in 1998 [41]. The intuition for this model comes from continuous time classical Markov chains. Essentially, the evolution of our walker on the graph is described by the Schrödinger equation with Hamiltonian defined as the adjacency matrix of said graph.

The continuous-time quantum walk takes place entirely in the position space $\mathcal{H}_p$ spanned by the positions of the particle. For a line with grid-length 1 this space is spanned by basis states $\{|i\rangle : i \in \mathbb{Z}\}$. Again, we will restrict ourselves to a finite line, so our space is spanned by the basis states $\{|i\rangle : i = 0, \dots, n\}$, $|i\rangle$ corresponding to a particle localized in position i in the chain. Unlike in the discrete case, no coin space is needed and no coin is flipped. So, let us define a graph G , with vertex set V . A step in the classical random walk can be described by a matrix M which transforms the probability distribution over V . The entries $M_{i,j}$ give the probability γ of going from i to j in one step of the walk. M is thus defined as

$$M_{i,j} = \begin{cases} -\gamma & \text{for } i \neq j, \text{ and } i \text{ and } j \text{ connected by an edge} \\ 0 & \text{for } i \neq j, \text{ and } i \text{ and } j \text{ not connected} \\ k\gamma & \text{for } i = j, \text{ where } k \text{ is the valence of vertex } i \end{cases} \quad (10)$$

This corresponds roughly to the adjacency matrix of the graph G .

Following the process defined by Farhi and Gutman [41], the Hamiltonian corresponding to the evolution of the CTQW is then defined as follows:

$$\langle i|\hat{H}|j\rangle = \begin{cases} -\gamma & \text{for } i \neq j, \text{ and } i \text{ and } j \text{ connected by an edge} \\ 0 & \text{for } i \neq j, \text{ and } i \text{ and } j \text{ not connected} \\ k\gamma & \text{for } i = j, \text{ where } k \text{ is the valence of vertex } i \end{cases} \quad (11)$$

Remark 3.3. The valence of a vertex i corresponds to the number of neighbors of this vertex.

We may then finally define a CTQW as follows:

Definition 2 (Continuous-time Quantum Walk). *A continuous time quantum walk over a graph G is defined by the unitary transformation $\hat{U} = e^{-i\hat{H}t}$ such that the state vector in the Hilbert space at a time t is given by the evolution $|\psi(t)\rangle = \hat{U}|\psi(0)\rangle$*

Testing our model through simulation in QuTiP Python library

Using Python's QuTiP library, we will simulate the time evolution of a the Hamiltonian defined for a continuous-time quantum walk. We are interested in learning more about the dynamics of excitation transfer through a chain of TLS.

A chain of 2 TLS is represented by the rather trivial adjacency matrix

$$M^{(2)} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad (12)$$

We may thus easily infer the form of the Hamiltonian under which the CTQW occurs: $\hat{H} = M$. We may then time-evolve our system under the Schrödinger equation, and find that the excitation harmonically oscillates back and forth from the first to the second site 7.

Let us now have a look at a slightly larger chain: that of 3 TLSs. We will also compare a chain with a loop (where nodes 1 and 3 are connected) 5. The adjacency matrices of these graphs take the respective forms:

$$M_{chain}^{(3)} = \frac{1}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \quad (\text{fig. 5a}) \quad \text{and} \quad M_{loop}^{(3)} = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} \quad (\text{fig. 5c}) \quad (13)$$

In both cases, the diagonal terms are null, because there is no chance of the excitation staying at its position after a time increment, i.e. no node is connected to itself. The only difference lies in some of the off-diagonal terms, which indicate the interaction between the limiting nodes in the graph—the boundary conditions.

Notice furthermore in fig. 6 that a chain with three qubits will behave in a very similar fashion to a loop with four graphs. This makes sense if one considers the symmetry of each object.

$$\begin{array}{ccc} & (1) & \\ \swarrow & & \searrow \\ (4) & & (2) \\ \searrow & & \swarrow \\ & (3) & \end{array} \quad \text{and} \quad (1) \leftrightarrow (2) \leftrightarrow (3) \quad (14)$$

We expect that the dynamics of going from (1) to (3) are the same in both cases, with different distributions.

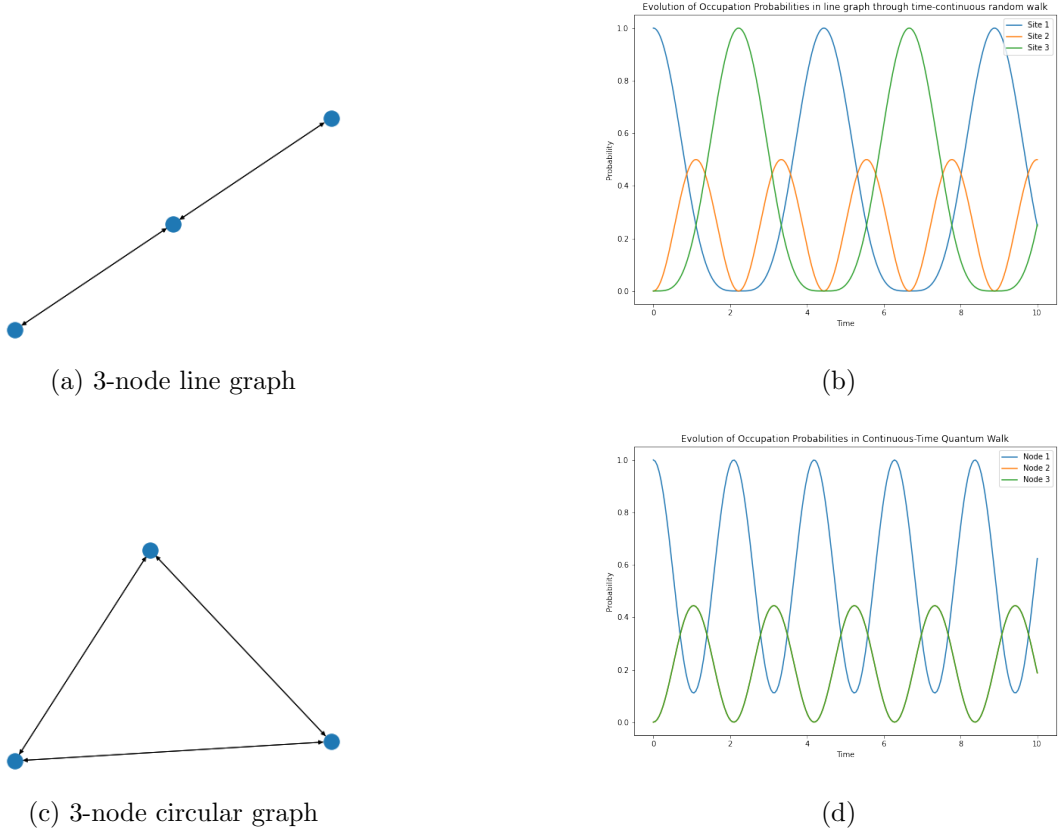


Figure 5: Evolution of occupation probability of excitation localization through CTQW in chain (loop) of 3 TLS. (**N.B:** In 5d, the green and orange lines overlap)

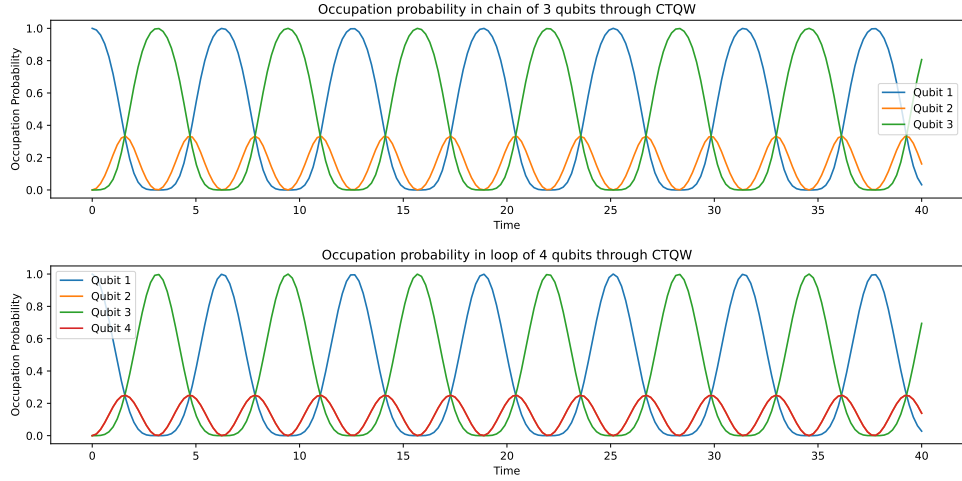


Figure 6: CTQW Occupation probability time evolution for TOP: Line-graph of 3 qubits BOTTOM: Loop-graph of 4 qubits (**N.B:** In the bottom graph, the red and orange lines overlap)

3.3 The classical alternative

Finally, let us compare the efficiency of this quantum walk with its classical alternative, a discrete random walk—otherwise known as incoherent hopping. In this case, energy hops from one site

to another at a fixed rate i.e., at each time step, the energy at each site is localized and has a certain probability of moving to a neighboring site. Our walker's position is initialized at 0 on the integer line $\mathbb{Z}$ restricted to integers smaller than n (length of our chain in other models). A fair coin is flipped. If the coin falls on heads, the walker will move to the right. Otherwise, it will move to the left. We set up a boundary condition such that, if the walker lands on positions 0 or $n - 1$, it must turn around. We implement this random walk in python $N = 1000$ times, and plot the average position of the walker at each time step in the following figure.

For a chain of length 2, the classical picture is fairly similar to the classical one. As we would expect, the walker arbitrarily chooses to go right or left so, on average, its position will harmonically oscillate.

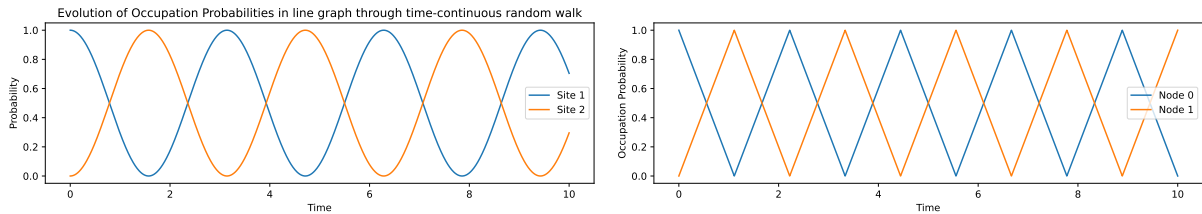


Figure 7: Time-evolution of occupation probabilities on 2-qubit graph, TOP: Classical random walk, Bottom: continuous-time quantum walk

For a longer chain, however, the picture is more complex. Indeed, comparing this classical walk with the quantum one, we can see that the end node in the line is reached much faster in the quantum setting. This is clear when looking, for instance, at the evolution of the occupation probability at being at the last node in the chain in the following figure.

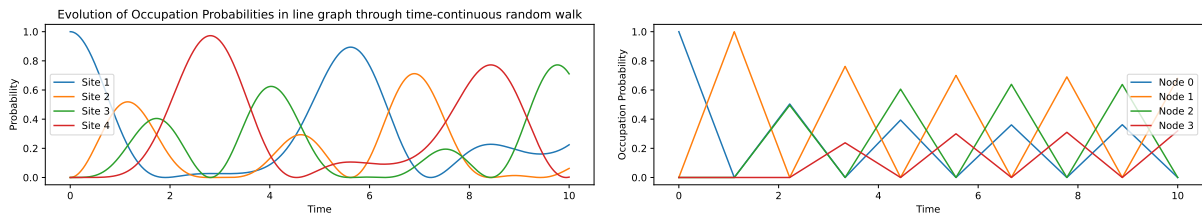


Figure 8: Comparing Quantum Walk with Classical Walk in 4 qubit chain

In the quantum case, the probability of reaching the end of the node rises quickly to near one whereas in the classical case, the probability of reaching the final node in the chain is an exponential function of the choices made by the “hopping” agent at each node.

3.4 Summing up

Ultimately, the Heisenberg model and quantum walk yield similar results, i.e. the occupation probabilities of being at each node in the graph evolved in the same way. In particular, in the case where $J = 1/2$ for the Heisenberg XY next-neighbor model, the simulations are identical. Indeed, it is consistent with existing literature, for instance with [42] to find that the action XY Heisenberg Hamiltonian on our state space is equivalent to the action of the adjacency matrix. This is indeed what we observe in figure 9

The models, however, are constructed very differently, as the former evolves in a 2^n dimensional Hilbert state whereas the later evolves in an n dimensional Hilbert space. This is because,

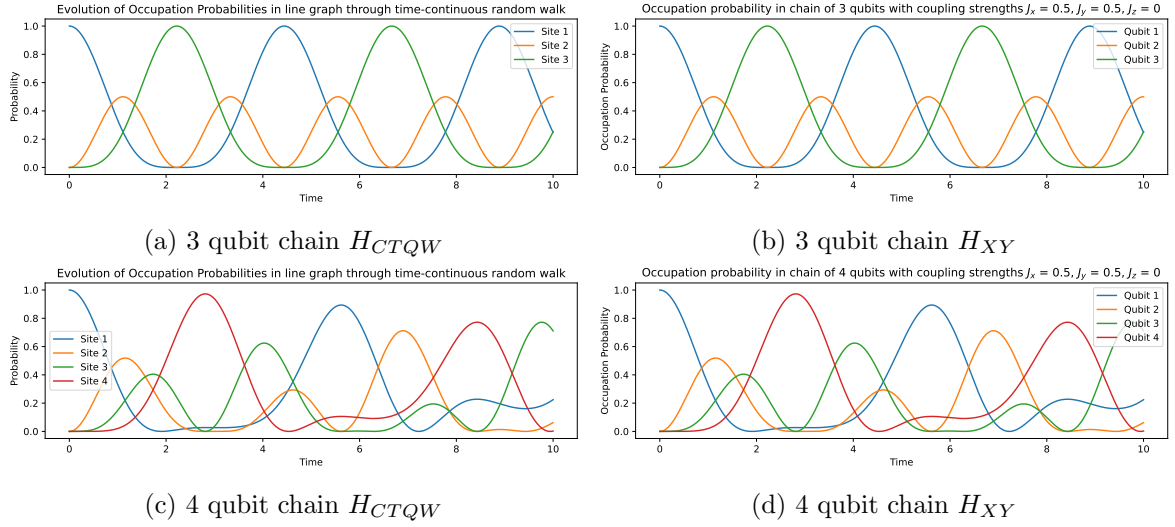


Figure 9: Comparing time-evolution of occupation probabilities in chain of 3 and 4 qubits for Hamiltonians H_{XY} with G_{CTQW}

in one case, we modeled our excitation as climbing the energy-level ladders of a single n -level system, whereas in the other, we considered it as jumping from one TLS to another. In fact, if we consider an initial state of the form $|0 \dots 1 \dots 0\rangle$, i.e. in which at any point in time only one TLS is excited, then the state transfer dynamics of our system are completely determined by the evolution in the N -dimensional subspace spanned by the basis vectors $\{|n\rangle, n = 1, \dots, N\}$ corresponding to spin configurations in which all spins are down apart from just one spin at the vertex n which is up.

Moreover, the crucial difference between the quantum and the classical walk are the quantum coherent dynamics, which are present only in the first case [40]. Whether or not such dynamics can occur is a result of the way systems in question are physically arranged. This includes the question whether excitation is shared across sites—and therefore delocalized. The direct comparison of diffusion rates between the quantum walk and classical walk also critically depends on the concrete parameters in these models, specifically the coupling strengths. Varvelo et al. [43] suggest that in light-harvesting systems found in nature such as the FMO complex of chromophore molecules the physical parameters happen to favor coherent dynamics—at least on a short range—allowing them to contribute to the observed high efficiency of energy transfer.

More physical limits to our representation of light-harvesting systems

Earlier in this chapter, we considered some physical limitations when setting up our light-harvesting system as a 1D chain of TLSs. In addition to these, we must point out that our model has only considered short chains. This may seem like a simplification but is indeed consistent with the actual structure of chromophore networks—for instance, the FMO complex is a 7-chromophore system [44]. Furthermore, we have ignored possible long-range interactions, and only considered next-neighbor coupling. This is left as future work, and seems an intriguing avenue to explore, as existing work has already shed light on the intrinsic tie between long-range interactions and transient delocalization [45].

This chapter has shown how under circumstances of sufficiently strong coupling, quantum dynamics can lead to a faster transfer of excitation than via classical hopping. Recent research

suggests that this distinction lies at the heart of the high efficiency of many light-harvesting systems found in nature.

Motivation for turning to adiabatic quantum computation

The quantum walk was introduced as a potential model for exciton diffusion in 1972. The formalism of the quantum walk was later defined by Fahri and Gutmann in 1998 [41]. In the following years, the same team developed an analogue framework for quantum computation, known as Adiabatic Quantum Computation [46]. Around this period, an increasing body of work showed the efficient nature of exciton transport in light harvesting systems, paying particular attention to the role of quantum effects such as delocalization and coherence. Around the same time, a growing body of work was focusing on quantum computing, and in particular focusing on the potential jump in efficiency of quantum computers as compared with their classical counterparts. In the face of such promise, it may have seemed like a natural next step for Farhi et al. to move from exploring the coherent transport of energy to the analogue transport of information. We will follow the same train of thought in our next section—remembering that information is, after all, physical and encoded in energetic states.

4 Quantum annealing

4.1 Adiabatic quantum computation

The concept of building quantum computers was initially ignited by physicist Richard Feynman during a talk where he proposed their use for simulating quantum mechanical processes. Shortly thereafter, David Deutsch introduced the quantum gate model: a universal framework for quantum computation [47]. Adiabatic quantum computation (AQC) is an alternative to this better-known gate model of quantum computation. Its foundations were laid by Edward Farhi and his collaborators in their seminal paper published in 2000. In this work, Farhi, Goldstone, Gutmann, and Sipser introduced the concept of utilizing adiabatic evolution for solving optimization problems. Research by Farhi and colleagues [46], as well as Aharonov and collaborators [39], demonstrated that these two models are polynomially equivalent. One distinction between the two models is their discrete versus analog natures.

The central tenet of AQC lies in the gradual transformation of a quantum system from an initial Hamiltonian to a final Hamiltonian. The initial Hamiltonian is chosen such that its ground state is straightforward to construct. Conversely, the final Hamiltonian encodes the solution to the optimization problem of interest. The key requirement is that this evolution occurs slowly enough to ensure that the system remains close to its ground state throughout the process.

4.2 The adiabatic theorem

The adiabatic theorem is the result at the heart of AQC. Its original formulation is attributed to Max Born and Vladimir Fock in 1928 [48]. In short, it tells us that a quantum system that begins in the non-degenerate ground state of a time-dependent Hamiltonian will remain in the instantaneous ground state, provided the Hamiltonian changes sufficiently slowly.

More formally:

Theorem 4.1 (Adiabatic Theorem). *Under a slowly changing time-dependent Hamiltonian $H(t)$ with instantaneous eigenstates $|n(t)\rangle$ and corresponding energies $E_n(t)$, a quantum system evolves*

from the initial state

$$|\psi(0)\rangle = \sum_n c_n(0)|n(0)\rangle$$

to the final state

$$|\psi(t)\rangle = \sum_n c_n(t)|n(t)\rangle,$$

where the coefficients undergo the change of phase

$$c_n(t) = c_n(0)e^{i\theta'_m(t)}e^{i\gamma_m(t)}$$

with the dynamical phase

$$\theta_m(t) = \frac{-1}{\hbar} \int_0^t E_m(t')dt'$$

and geometric phase

$$\gamma_m(t) = i \int_0^t \langle m'(t') | m'(t') \rangle dt'.$$

In particular, $|c'_n(t)|^2 = |c_n(0)|^2$, so if the system begins in an eigenstate of $H(0)$, it remains in an eigenstate of $H(t)$ during evolution with a change of phase only.

Although this is beyond the scope of this thesis, an insightful demonstration of the theorem can be found in Andrew M. Childs' Lecture Notes on Quantum Algorithms [49].

A note on running time and spectral gap conditions

The success of QA is determined by how slowly the Hamiltonian changes with time, and on the spectral gap $\delta_m = \min_{t < t_f} \Delta(t) = \min_{t < t_f} |g(t) - 1(t)|$ between the ground state and first excited state of the instantaneous Hamiltonian. In sum, if

1. the register Q of the n-qubits in our system is in ground state at $t = 0$
2. δ_m is strictly greater than 0
3. the process evolves slowly enough

then the adiabatic theorem will hold.

More specifically, the criterion of the adiabatic time evolution [50] is given by

$$\max \left(\frac{|1(t)| \frac{dH(t)}{dt}}{g(t)} \right) \ll 1, \min [\Delta(t)^2] \quad (15)$$

where:

1. $g(t)$ and $|1(t)|$ represent the instantaneous ground and first-excited states at time (t), respectively.
2. $\Delta(t)$ denotes the instantaneous energy gap above $g(t)$.

Roughly speaking, this expression suggests that Quantum Annealing (QA) works better for larger $\Delta(t)$.

4.3 Defining quantum annealing

The terms “Adiabatic Quantum Computation” and “Quantum Annealing” are often used interchangeably, although they have distinct meanings. We will focus on the latter in this thesis, so it is useful to shed some light on their differences. Quantum Annealing (QA) describes a type of heuristic search algorithm that can be implemented on an AQC platform. In other words, because it is impossible for any real-world computation to run in perfect isolation, QA may be thought of as the real-world counterpart to AQC, a theoretical ideal.

An AQC algorithm to solve optimization problem P with objective function h works on a register Q of n qubits having superposition state $|\psi_t\rangle$ at time t . The algorithm is described by a time-varying Hamiltonian $H(t)$, specified by three components:

1. An initial Hamiltonian H_0 chosen so that the ground state of the system is easy to find
2. A problem Hamiltonian H_P that encodes the objective function so that the ground state of Q is an eigenstate of H_P having minimum eigenvalue. That is, a ground state s_g of Q corresponds to an optimal solution to our problem
3. An adiabatic evolution path function $s(t)$ that decreased from 1 to 0 as $t : 0 \rightarrow t_f$ (t_f being the final time). For now we use a simple linear path $s(t) = 1 - t/t_f$

Many well-known and difficult-to-solve optimization problems can be cast as problems of minimization of objective cost functions. Naturally, the task of finding the minimum of a cost function can be cast to the quantum mechanical problem of finding the ground state of a Hamiltonian. Given a cost function $h : z \rightarrow h(z)$ to minimize, we will define a cost Hamiltonian H_C :

$$H_C := \sum_{z \in \{0,1\}^n} h(z) |z\rangle\langle z| \quad (16)$$

The correspondence between the objective minimum of h and the ground state of H_C is clear. Hence, the ground state of H_C is of interest as it encodes the solution to our optimization problem or the minimum of the objective function of interest. In general, finding this ground state is not a trivial task, which is where quantum annealing comes in.

Remark 4.2. Note that the problem Hamiltonian of interest H_C is diagonal in the computational basis generated by $\{|z\rangle\langle z|, z \in \{0,1\}^n\}$. This is coherent, because the final ground state we are interested in finding should be a classical state.

Defining the problem Hamiltonian

In general, defining H_P is not a trivial task, and the form we look to cast it on will largely depend on the hardware of the quantum annealer. In this report, we’ll focus on a problem formulation in line with the methods presented by D-Wave (see section 4.4 for more details). Thus, transformation of a general optimization problem instance to a native instance is a two-step process: first, translate the problem to a Binary Quadratic Model (BQM), and second, map the general BQM connectivity graph to the hardware working graph.

Binary Quadratic Models - Ising & Quantum Unconstrained Binary Optimization (QUBO) models

In general, quantum annealing is fit for solving BQM problems, which consist of Ising models and quadratic unconstrained binary optimization (QUBO) models. This is a generalization of the native problem that D-Waves quantum annealing chips are designed to solve [51].

An Ising Model problem takes the form

$$\min \left(\sum_i h_i s_i + \sum_i \sum_{j>i} J_{i,j} s_i s_j \right) \quad (17)$$

where the binary variables s_i and s_j may take values from $\{-1, 1\}$.

A QUBO model takes the form

$$\min \left(\sum_i a_i x_i + \sum_i \sum_{j>i} b_{i,j} x_i x_j + c \right) \quad (18)$$

where the binary variables x_i and x_j may take values from $\{0, 1\}$.

It turns out that these two forms are mathematically equivalent, and that one may easily map one to the other thanks to the two equivalent substitutions:

$$x_i \mapsto \frac{s_i - 1}{2} \text{ or } s_i \mapsto 2x_i + 1 \quad (19)$$

Remark 4.3. One sometimes encounters this mapping as inverted in the literature, i.e. a ‘+’ instead of a ‘−’ and vice versa. This depends on the way the system is defined. In our case, we define the basis states of each TLS as

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (20)$$

because this is the convention in Python’s QuTiP library, which we have used to simulate our annealing process.

The initial hamiltonian H_0

Many (actually infinite) choices are possible for defining our initial Hamiltonian. It is only essential that our choice respect two constraints:

1. The ground state of H_0 ought to be *easy* to prepare
2. H_0 should not commute with the problem Hamiltonian at hand

The first condition is fairly trivial as it is mentioned in the definition of the QA algorithm. If the second condition is not met, i.e. if $[H_0, H_P] = 0$, and the system was initially prepared in the ground state of H_0 , then there would be no time evolution. Indeed, if two operators commute, they share the same eigenvectors.

A natural choice for H_0 is

$$H_0 = - \sum_{j=1}^n \sigma_x^{(j)} \quad (21)$$

Whose ground state is known as a superposition of all possible states:

$$|S\rangle := \frac{1}{\sqrt{2^n}} \sum_{z \in \{0,1\}^n} |z\rangle \quad (22)$$

The annealing schedule

Finally, for any given H_0 and H_P , there are many possible choices for the interpolation $H(t)$. One simple choice we will make is a time-dependent Hamiltonian of the form

$$H(t) = \left(1 - \frac{t}{t_f}\right) H_0 + \frac{t}{t_f} H_P \quad (23)$$

4.4 Physical implementation of QA

The main commercial player in QA today is D-Wave Systems. They manufacture computer systems that implement a quantum annealing algorithm in hardware. Our optimization problem, which was mapped to an IM problem, must be translated to an equivalent problem that matches the connectivity structure of the Quantum Processing Unit (QPU). In current QPU models, qubits and couplers are structured according to a Chimera graph. The superconducting QPU at the heart of the D-Wave system, which operates at a temperature of approximately 12 mK, is a controllable, physical realization of the quantum Ising spin system in a transverse field. An in depth look at this architecture goes somewhat beyond the scope of this thesis, but I invite the curious reader to find out more about D-Wave's QPU architecture from their website.

The Hamiltonian used by D-Wave in quantum annealing takes the form of the IM:

$$H_{Ising} = \frac{-A(s)}{2} \left(\sum_i \sigma_x^i \right) + \frac{B(s)}{2} \left(\sum_i h_i \sigma_z^i + \sum_{i>j} J_{i,j} \sigma_z^i \sigma_z^j \right) \quad (24)$$

Roughly speaking, D-Wave systems always choose the initial Hamiltonian to be of the form 21, which is sometimes called a transverse field Hamiltonian. Initially, each qubit and coupler are programmed to have a coefficient of 0, which are then increased according to the QPU's allowed values to introduce the problem Hamiltonian.

5 Simulating quantum annealing: Shortest path problem

5.1 The shortest path problem

We are interested in learning more about the role of delocalized states in quantum annealing. The best way to gain an understanding of this is by simulating a concrete example of an annealing process and observing the behavior of these states. Let us recall the broad scope of the thesis, which invites us to set up a model that allows us to compare the processes of exciton diffusion with quantum annealing. Because the diffusion of excitons is essentially the process of an excitation transferring from a starting point to an ending point in a lattice, the choice of solving the shortest path problem is a natural one.

Formulating the problem

Let us consider that we are given a graph $G = (V, E)$, where V is the set of the graph's vertices and E is the set of the graph's edges, i.e.:

$$V = \{v_1, \dots, v_{|V|}\}$$

$$E = \{< i, j > \text{ such that } v_i \text{ and } v_j \text{ are connected in the graph}\}$$

We may define the entries of the graph's adjacency matrix A , such that:

$$A_{ij} = \begin{cases} d_{i,j} & \text{if } < i, j > \in E \\ 0 & \text{otherwise} \end{cases} \quad (25)$$

For the sake of simplifying the model, we will only consider unweighted graphs, in which all the edges are equidistant, so that $d_{i,j} = 1 \forall i, j$.

Problem: Find the shortest path from a starting vertex v_s to a terminal vertex v_t .

Let us map our system to a register Q of $|V| = n$ qubits. Each TLS will represent a vertex on the graph. In order to mathematically formulate the problem, we define a binary variable

$$z_i = \begin{cases} 1 & \text{if } v_i \text{ is in the path} \\ 0 & \text{otherwise} \end{cases} \quad (26)$$

In other words, if a vertex v_i is on the path, the associated i^{th} qubit will be in an excited state. Hence we represent a given path in the Hilbert space $\mathcal{H} = \bigotimes_{i=1}^{|V|} \mathbb{C}^2$ by the state:

$$|z_1\rangle \otimes |z_2\rangle \otimes \dots \otimes |z_{|V|}\rangle = |z_1 \dots z_{|V|}\rangle \quad (27)$$

Hence, the cost function we need to minimize is:

$$h(\underline{z}) = \sum_{i=1}^{|V|} z_i \quad (28)$$

5.2 Defining the problem Hamiltonian

The goal of this section is to define the problem Hamiltonian H_P , whose ground state should minimize the cost function h associated to our shortest path problem.

Of course, it is not possible to simply read off the z_i terms in $\underline{z}$. Instead, we need to define an operator, whose eigenvalues will give us equivalent information. Thankfully, there exist a common set of operators known as the Pauli group which give us such information on a state. In particular, the Pauli Z operator, which is sometimes referred to as a bit flip, will come in handy. Recall a fundamental property of this operator:

$$\sigma_z(\alpha|0\rangle + \beta|1\rangle) = \alpha|0\rangle - \beta|1\rangle \quad (29)$$

Hence, it is possible to define our desired operator

$$\left(\frac{1 - \sigma_z}{2}\right)(\alpha|0\rangle + \beta|1\rangle) = \frac{1}{2}(\alpha|0\rangle + \beta|1\rangle) - \frac{1}{2}((\alpha|0\rangle - \beta|1\rangle)) = \beta|1\rangle \quad (30)$$

More generally, for a state $|\underline{z}\rangle = |z_1 \dots z_{|V|}\rangle$, we will define the Pauli Z operator acting only on the i^{th} qubit as

$$\sigma_z^i = \mathbb{1} \otimes \dots \otimes \underbrace{\sigma_z}_{i^{th} \text{ term}} \otimes \dots \mathbb{1}$$

So, combining equations 16 and 28 we may easily define the corresponding Hamiltonian, whose ground state encodes the minimum of our cost function:

$$H_C := \sum_{\underline{z} \in \{0,1\}^n} h(\underline{z}) |\underline{z}\rangle \langle \underline{z}| = \sum_{i=1}^{|V|} \frac{1 - \sigma_z^i}{2} \quad (31)$$

It is rather trivial to express the ground state of H_C : $|\psi_{ground}\rangle = |0\rangle^{\otimes n}$. In *graph* terms, this path is indeed the shortest, but it is also impractical as it goes through no vertices! It is thus clear that we need to impose some additional constraints on our system.

We will formulate the constraints of our problem as follows:

1. The path should start at a vertex v_s

$$H_s = \frac{1 - \sigma_z^s}{2} \quad (32)$$

2. The path should end at a vertex v_t

$$H_t = \frac{1 - \sigma_z^t}{2} \quad (33)$$

3. The path should be continuous, so that every vertex (except for v_s and v_t), should be a part of exactly 2 edges

$$H_E = \sum_{i=1 \notin \{s,t\}}^n 2 - \sum_{j=1}^n A_{ij} \frac{(1 - \sigma_z^i)(1 - \sigma_z^j)}{4} \quad (34)$$

Proof. Let us briefly show (in a quasi-proof) that each one of these terms does it job:

1. Let $|\psi\rangle = |z_1 \dots z_n\rangle$, then

$$\hat{H}_c |\psi\rangle = \frac{1 - \sigma_z^s}{2} |\psi\rangle = \frac{|\psi\rangle}{2} - \frac{1}{2} \sigma_z^s |\psi\rangle = \begin{cases} \frac{1+1}{2} |\psi\rangle = |\psi\rangle & \text{if } z_s = 1 \\ & \text{i.e. if } v_s \in \text{path} \\ \frac{1-1}{2} = 0 & \text{otherwise} \end{cases}$$

2. The proof is identical for $\hat{H}_t$

3. To understand how H_E acts, let us focus on how this operator acts on a state $|\psi\rangle =$

$|\dots z_i \dots z_j \dots\rangle$, without loss of generality

$$\begin{aligned}
A_{ij} \frac{(1 - \sigma_z^i)(1 - \sigma_z^j)}{4} |\psi\rangle &= \begin{cases} A_{ij} |\psi\rangle & \text{if } z_i = z_j = 1 \\ 0 & \text{if } z_i \neq z_j \text{ or if } z_i = z_j = 0 \end{cases} \\
&= \begin{cases} |\psi\rangle & \text{if } z_i = z_j = 1 \text{ and } \langle i, j \rangle \in E \\ 0 & \text{if } z_i \neq z_j \text{ or if } z_i = z_j = 0 \text{ or if} \\ & \text{or if } A_{ij} = 0 \text{ i.e. if } \langle i, j \rangle \notin E \end{cases}
\end{aligned}$$

Hence, summing over all values of j , we count the number of *legal* edges in the path represented by the state $|\psi\rangle$. The total Hamiltonian's energy will be minimized when this sum is most often equal to exactly 2, i.e. when each vertex in the path is connected to exactly 2 other vertices, except for v_s and v_t , which should only have one connection.

□

Equipped with these terms, we can formulate our final problem Hamiltonian. We will assign a weight to each penalty term, dependent on the problem.

$$H_P = \alpha_C H_C + \alpha_s H_s + \alpha_t H_t + \alpha_E H_E \quad (35)$$

Choosing our penalty constants

There is no universal way of choosing these penalty factors, and this is generally done through trial and error. However, choosing the wrong scaling factors may result in finding final ground states which do not meet the constraints. Often, the ratio of these penalty factors is found considering the worst case scenario [52]. For instance, in the case of the shortest path problem, one could imagine a limiting case in which the shortest path found by our cost Hamiltonian is $|0 \dots 0\rangle$, and no constraints are met. Some work has been done on finding penalty constraints for well-known optimization problems [52, 53, 54]. However, to the best of my knowledge, this has not been done for the shortest path problem, we hence have to determine these by ourselves!

To set our penalty constants, we start by setting $\alpha_C = 1$, and adjust the other terms accordingly. Let $C = \sum_{i,j=1}^{|V|} A_{ij}$, with A the adjacency matrix of our graph G . through trial and error, we found that we obtain a reasonably valid solution by taking $\alpha_s = \alpha_t = C$ and $\alpha_E = 1/C$. Hence the final problem Hamiltonian corresponding to our problem is

$$\begin{aligned}
H_P &= H_C + C H_s + C H_t + \frac{1}{C} H_E \\
&= \sum_{i=1}^n \frac{1 - \sigma_z^i}{2} + C \left(\frac{1 - \sigma_z^s}{2} + \frac{1 - \sigma_z^t}{2} \right) + \frac{1}{C} \left(\sum_{i=1 \notin \{s,t\}}^n 2 - \sum_{j=1}^n A_{ij} \frac{(1 - \sigma_z^i)(1 - \sigma_z^j)}{4} \right) \quad (36)
\end{aligned}$$

Furthermore, this is consistent with the penalty factors found in other works related to the similar Travelling Salesman Problem TSP, such as [52, 54, 53]

5.3 Testing our model through simulation using the Qutip Python library

The development of quantum annealing was motivated by the prospect of solving complex optimization problems. Within the frame of this thesis, we are looking to gain a deeper understanding of how annealing works by solving a “simple” problem, for which some classical solutions already exist, and are well documented [55]. The point here is not to demonstrate any form of quantum advantage, but to study the underlying mechanisms. To gain some intuition about the abstract problem Hamiltonian with which we are faced, let’s see how it would act on a simple system. We should first consider the trivial 1 and 2-vertex graphs but will skip these because they do not offer enough insight into the energy landscape of the problem Hamiltonian at hand. Instead, let us consider a small 4-vertex graph, as illustrated in figure 10.

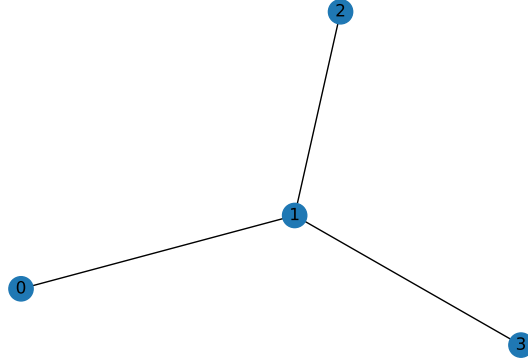


Figure 10: Schematic of simple example of 4-vertex graph

The problem Hamiltonian here will take the form:

$$\hat{H}_P = \sum_{i=1}^4 \frac{1 - \sigma_z^i}{2} + 4 \left(\frac{1 - \sigma_z^s}{2} + \frac{1 - \sigma_z^t}{2} \right) + \frac{1}{4} \left(\sum_{i=1 \notin \{s,t\}}^4 2 - \sum_{j=1}^4 A_{ij} \frac{(1 - \sigma_z^i)(1 - \sigma_z^j)}{4} \right) \quad (37)$$

The energy spectrum of this problem Hamiltonian is relatively easy to analyze because all terms are diagonal in the computational basis, and therefore commute. This allows us to determine the quality of the anneal and play with the deciding parameters: namely t_f .

To better understand the role of each part of the problem Hamiltonian, it is thus insightful to take a look at their respective Hinton diagrams in the case where $|V| = n = 4$, and set $s = 0$, $t = n - 1 = 3$.

Let’s check in these energy landscapes that the cost and penalty Hamiltonians we defined are doing their jobs. The Hinton diagrams of H_s and H_t are the easiest to verify. It is clear from 11c that the energy of the states in which the first qubit is excited, i.e. the path goes through the first vertex, are given lower energy. The same can be said as a consequence of H_t 11d, wherein the states ending on the last vertex have lower energy. The cost function Hamiltonian H_C 11a seems to be correctly allocating to each state an energy corresponding to the number of vertices it represents. Finally, the edge penalty Hamiltonian H_E 11b gives a higher energy to the states with illegal (ie not in E) paths or too few paths while giving lower energy to states with the right number of legal connected paths.

Let us now see how well our quantum annealing simulation will perform. It is clear from figure 12 that the ground state (smallest eigenvalue) of H_P corresponds to the state $|1101\rangle$.

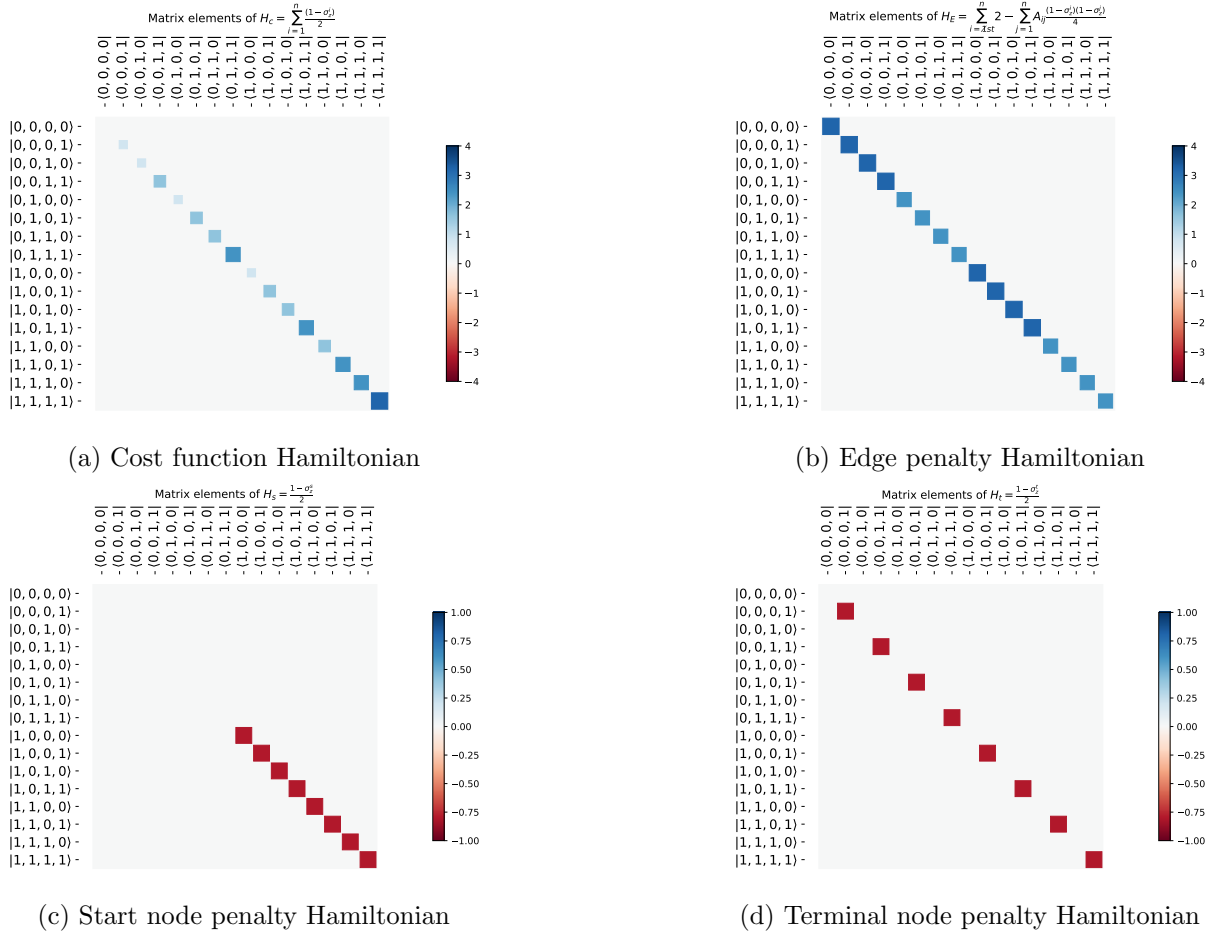


Figure 11: Hinton diagram for each subset of the problem Hamiltonian - related to graph [10]

Coincidentally, this corresponds to what we would expect to find as a shortest path through the four node graph 10 i.e. $0 \rightarrow 1 \rightarrow 3$.

Notice, however, that the spectral gap between this energy and the one of the state $|1111\rangle$ is quite small. Hence, according to the criteria laid out in equation 15, the QA process should work better for larger spectral gaps. As a consequence, we may expect that if the annealing time t_f is chosen to be too small, our system would remain trapped in a local minimum energy. Indeed, figure 13 clearly shows that the quality of the anneal will decrease if t_f is chosen to be too small. Figure 13d shows that if for instance $t_f = 1$, the system will not end up in the group state of H_P , but rather in superposition of possible eigenstates of H_P . The graph also clearly shows that the system is initialized in the ground state of H_0 , an equal superposition of all basis states.

5.4 Summing up

In this section, we went over the formalism for defining an optimization problem that may be solved by quantum annealing and we implemented a case study by looking at the example of the shortest path problem.

Within the frame of this thesis, we were interested at comparing energy transport with

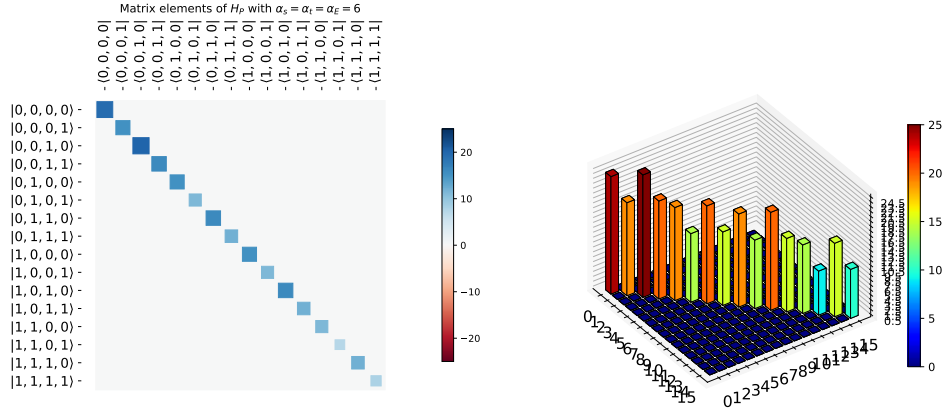


Figure 12: Hinton diagram and Histogram of H_P for 4-vertex graph

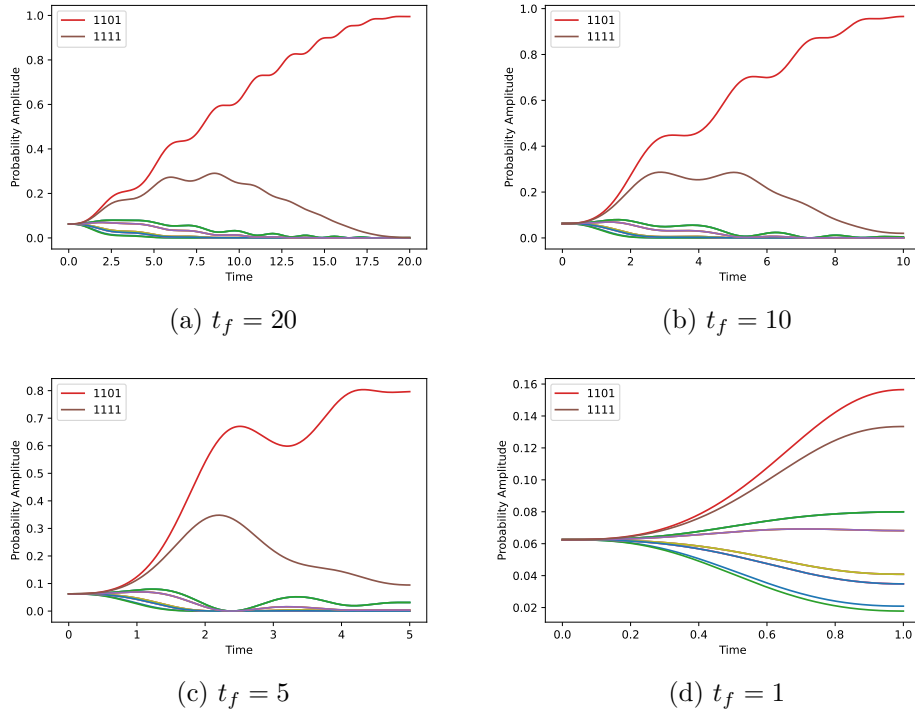


Figure 13: Probability amplitude plot QA through 4-Node graph

information transport in quantum systems that involve delocalized states and coherent dynamics. QA is a clear example of how physical processes can facilitate information processing. Overall, QA is an analog model for quantum computation, during which a gradual transformation is made from an initial Hamiltonian to a final Hamiltonian. This transformation/annealing process, is where delocalized states come into play. Natural laws dictate that systems tend to put themselves in the lowest energy configuration, and the adiabatic system allows us to use this property to our advantage. The key is then to encode our problem into an energy landscape, whose minimum would be difficult to find explicitly—this is the challenge with QA.

Limits of our analysis

Much like in our model for exciton diffusion, we made many simplifications when setting up the shortest path problem: assuming that we would only need to deal with simple graphs, with uniform weight 1. Moreover, as we were restricted by the computational power of a personal computer, we were not able to simulate the anneal for larger, more complex, configurations. Hence, there are many interesting avenues along which this work could be extended. One may, for instance, want to take an experimental approach to the problem and implement the anneal on, for instance, D-Wave hardware, to observe its efficacy—and consider more complex configurations. Furthermore, one may be interested in quantifying the use of quantum resources in the QA process, as a potential proof (or disproof) of quantum advantage.

6 Comparing

Let us now take a closer look at comparing the proposed models for exciton diffusion and quantum annealing.

6.1 Comparing the models

Random walks

Throughout this thesis, the points of comparison between exciton diffusion and QA arose quite naturally. From a historical perspective, both processes fit within a similar framework which suggested the use of random walks in continuous time quantum processes. On one hand, we found that the diffusion of excitons in light-harvesting systems is described by a continuous-time random walk. On the other, QA can sometimes be imagined as a random walk taking place on a changing problem landscape. Because these processes follow similar time-evolution mechanisms—namely referring to the random walk process, the Hamiltonians at play are somewhat similar. Let us take a closer look at these.

A closer look at respective Hamiltonians

The most striking similarity between the descriptions of exciton diffusion in light-harvesting systems and quantum annealing is in their Hamiltonians. Both models (in a certain sense) involve an Ising model, with next neighbor coupling.

In the first part of this thesis, we went over various models for energetic state transfer through a 1D chain of TLSs, namely the Heisenberg next-neighbor model, and CTQW model, whose respective Hamiltonians were of the form:

$$\hat{H}_{XY} = J \sum_{i=1}^{N-1} (\sigma_x^{(i)} \sigma_x^{(i+1)} + \sigma_y^{(i)} \sigma_y^{(i+1)}) \quad (38)$$

$$\langle i | \hat{H}_{\text{CTQW}} | j \rangle = \begin{cases} -\gamma & \text{for } i \neq j, \text{ and } i \text{ and } j \text{ connected by an edge} \\ 0 & \text{for } i \neq j, \text{ and } i \text{ and } j \text{ not connected} \\ k\gamma & \text{for } i = j, \text{ where } k \text{ is the valence of vertex } i \end{cases} \quad (39)$$

In the simplest—if not trivial—case of a chain of two TLSs (which we compared in figure 7), we may explicitly write these Hamiltonians as:

$$\hat{H}_{XY}^{(2)} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 2J & 0 \\ 0 & 2J & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \text{ and } \hat{H}_{\text{CTQW}} = \begin{pmatrix} k\gamma & -\gamma \\ -\gamma & k\gamma \end{pmatrix} \quad (40)$$

The different dimensions of the respective Hamiltonians are a consequence of the different models used. Recall that, in the case of the Heisenberg next-neighbor interaction model, our $\mathcal{H} = \bigotimes_{i=1}^2 \mathbb{C}^2$ and is hence spanned by the basis states:

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$$

Whereas, in the CTQW, our quantum walk is taking place in the position space $\mathcal{H}_p$ spanned by the positions of the particle. In this particular case, this space is spanned by basis states

$$\{|0\rangle, |1\rangle\}$$

In our case-study of QA, which consisted in finding the shortest path through a graph, the problem Hamiltonian for a simple two-node graph takes this form (set $s = 1$, $t = 2$, $C = 2$):

$$\begin{aligned} H_P &= H_C + CH_s + CH_t + \frac{1}{C}H_E \\ &= \sum_{i=1}^2 \frac{1 - \sigma_z^i}{2} + 2 \left(\frac{1 - \sigma_z^1}{2} + \frac{1 - \sigma_z^2}{2} \right) + \frac{1}{2} \left(\sum_{i=1 \notin \{s,t\}}^2 2 - \sum_{j=1}^2 A_{ij} \frac{(1 - \sigma_z^i)(1 - \sigma_z^j)}{4} \right) \\ &= 3 \left(\frac{1 - \sigma_z^1}{2} \right) + 3 \left(\frac{1 - \sigma_z^2}{2} \right) \\ &= \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 6 \end{pmatrix} \end{aligned} \quad (41)$$

It is clear that the ground state of this Hamiltonian corresponds to the state $|11\rangle$, which is indeed the obvious shortest path through the graph. Our problem Hamiltonian is, as you will recall, a diagonal operator, as its eigenstates must be classical. On the contrary H_0 is only composed of off diagonal terms.

$$H_0 = - \sum_{j=1}^2 \sigma_x^{(j)} = -\sigma_x \otimes I - I \otimes \sigma_x = (-1) \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix} \quad (42)$$

Hence the total QA Hamiltonian is of the form

$$H(t) = \left(1 - \frac{t}{t_f}\right) H_0 + \frac{t}{t_f} H_P = \left(\frac{t}{t_f} - 1\right) \begin{pmatrix} 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{pmatrix} + \frac{t}{t_f} \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 6 \end{pmatrix} \quad (43)$$

Overall, although they are not identical, both the Hamiltonian corresponding to the next-

neighbor coupling in a light-harvesting system and the Hamiltonian representing the energy levels of our problem landscape share a common feature: off-diagonal terms. These terms are a hallmark of delocalized states and key determinants of their time evolution. Roughly speaking, the diagonal elements of a Hamiltonian correspond to the energy levels of the system (eigenvalues), whereas its off-diagonal elements connect different basis states. Indeed, in both systems, the TLSs of our system are coupled.

6.2 Comparing the physical systems

Light-harvesting complexes (LHCs) and quantum annealers represent intriguing parallels between energy and information transport and processing. Both LHCs and quantum annealers involve interactions among their fundamental constituents i.e. next-neighbor (or further range) interactions. In LHCs, these constituents are pigments (such as chlorophylls), while in QA, they are qubits. We viewed these pigments as TLSs for the sake of simplicity, but this is, of course, a simplification. Nevertheless, a study of existing literature proved the importance of coherent dynamics in both processes. In LHCs, delocalized excitons move through pigment arrays, efficiently transferring absorbed solar energy. Quantum annealers exploit adiabatic evolution to find optimal solutions by evolving states coherently. Finally, both systems are constructed (in the case of LHC, we might use the term naturally-selected or evolved instead)—whether it be by man or nature—to solve a kind of optimization process. In the case of LHC: to transfer energy efficiently across a landscape of sites. In the case of QA: to solve combinatorial optimization problems, seeking optimal configurations across a landscape of sites.

Throughout this thesis, we paid special attention to the role of delocalized states in both systems. Within light-harvesting complexes, excitons explore multiple pathways simultaneously. As photons are absorbed by pigments, excitons delocalize across neighboring pigments. This coherent exploration allows efficient energy transfer toward the reaction center, where charge separation occurs. The simultaneous exploration of various paths ensures robustness against environmental fluctuations and enhances overall energy capture. Quantum annealers, too, explore multiple paths concurrently. During the adiabatic evolution, the system explores a superposition of states, akin to a quantum walk. The qubits explore different configurations, seeking the lowest-energy solution. By traversing various paths simultaneously, quantum annealers enhance the chances of finding optimal solutions in complex optimization landscapes.

The main difference between both physical models is their sensitivity to noise. LHCs operate at ambient room temperature within living organisms. In some cases, their performance can even be enhanced by certain types of noise [56]. Quantum annealers, however, require cryogenic temperatures (near absolute zero) to maintain qubit coherence. This extreme cooling ensures minimal decoherence during quantum computations.

7 Discussion

Now that we have a better understanding of the points of comparison between delocalized states in both quantum annealing and exciton diffusion in light-harvesting systems, let us take a step back, to situate the relevance of this work. In particular, I would like to invite the reader to reflect on the evolution of scientific knowledge and how it affects many societal questions such as those related to technology and innovation, science funding and workforce development. Our comparison provide a powerful lens to consider such issues.

The journey from scientific discovery to technological innovation is often long and winding,

marked by significant delays between breakthroughs in the laboratory, conceptual insights, and mathematical tools on the one hand, and their practical applications in the wider world on the other hand. The concept of **path dependency** is crucial here, suggesting that early scientific discoveries and technological innovations can significantly influence the direction of future research and development efforts [57]. For instance, the invention of the transistor at Bell Labs, not only revolutionized electronics but also set the stage for the development of modern computing and digital communication, shaping subsequent research priorities and funding flows.

Throughout this thesis, we shed some light on the historical and conceptual interplay between quantum annealing and exciton diffusion. In section 3.4 in particular, we illustrated the cross-fertilization between the study of quantum walks and the development of adiabatic quantum computing. Continuous-time quantum walks, which model the quantum mechanical equivalent of classical random walks, have provided profound insights into the mechanisms of energy transport in light-harvesting complexes. Similar insights have, in turn, informed the development of adiabatic quantum computing, where the gradual evolution of a quantum system is harnessed to solve optimization problems. This synergy highlights the value of interdisciplinary approaches in science, demonstrating how theoretical advances in one area can spur innovation and progress in another.

Nevertheless, the path of scientific progress is not always smooth or predictable. For instance, the case of quantum computing and information processing starkly contrasts with the long-standing controversy over the role of coherent transport in photosynthesis. For years, the latter remained heavily debated. It was not until the publication of certain models and experimental results, such as discussed in the 2021 Varvelo paper [43], that strong support emerged for the role of quantum coherence in biological energy transport. The prolonged lack of insight and consensus meant that the use of delocalized states in energy applications remained more or less a niche topic over many years. This divergence in research trajectories underscores the inherently uncertain nature of scientific exploration, where evidence, consensus, and external factors such as societal needs and commercial interests interplay to determine the focus and funding of research endeavors.

Quantum computing too faced much controversy in its early days, but by the 2000s, a strong foundation of basic, widely agreed upon principles had emerged. By the early 2020s, this had spawned a new industry and, as such, has attracted a significant portion of the quantum engineering talent pool [58].

Such dynamics in the evolution of knowledge and in the emergence of sub-fields—even as they build on similar physical principles—raises important questions with respect to science and innovation policy. While emerging scientific knowledge can be viewed as the supply side of new technology, societal needs can be viewed as the demand side [59]. While scientific progress is in many ways an organic process that results from the decentralized activities of many actors, including accidents and coincidences, there are also controllable boundary conditions that affect this process. Policy makers, as representatives of society, want to ensure that new knowledge can be directed toward pressing problems of the time, and that imbalances, e.g. in the flow of talent, can be mitigated.

This concentration of resources and talent in quantum computing as one area of quantum engineering raises important questions about the sustainability and diversity of the broader quantum research ecosystem. The potential brain drain from other vital areas of quantum research necessitates a thoughtful policy response, aimed at ensuring a balanced distribution of talent and resources across the spectrum of potential technological applications inspired by quantum phenomena. Governments and institutions could play a pivotal role in crafting initiatives that

encourage diversity in research focus, supporting emerging areas of quantum science alongside established ones. In view of concerns about climate change, this could include the promotion of quantum engineering for energy applications, e.g. toward the development of high-efficiency organic solar cells.

Overall, the comparative analysis of quantum annealing and exciton diffusion demonstrates how the same physical principles—delocalized states and their coherent evolution—can manifest in different systems with different purposes; and as such inform different use cases and potential technologies. This underscores the importance of keeping disciplinary boundaries permeable in scientific research. and the relationship between organic evolution and deliberate boundary setting in science and science policy. It also acknowledges the interplay between theoretical discovery, practical application, and historical context.

8 Conclusion and outlook

The close connection between energy and information transport and processing has been pointed out before [60]. In fact, in the current and historical technology landscape, many information-handling devices are used otherwise for energy-transfer and vice-versa. The transistor epitomizes this interplay, as an information storage device strongly reliant on energy transfer for signal amplification. Another example is the capacitor, whose basic principle is deployed for information storage applications in dynamic random access memory (DRAM) chips, and for energy storage applications in so-called supercapacitors (SCs). It is natural then, in the dawn of the emergence of quantum information science (recall the vast investments being made in QT [1]), that we should investigate such parallels (see also X et al. 2023 [61]). Moreover, the topic of this thesis arose organically, as the study of quantum effects in exciton diffusion and model for AQC arose together. Indeed, around the year 2000, quantum walks were proposed as an alternative to the quantum Fourier transform in quantum algorithms. In parallel, groundwork was laid for adiabatic quantum computation. The timeline went as follows:

2000: Quantum computation by adiabatic evolution [46]

2003: Exponential algorithmic speedup by a quantum walk [62]

Overall, the parallels between energy and information transport uncovered in this thesis underscore the profound interconnectedness of quantum phenomena in diverse systems. By elucidating the role of delocalized states in both quantum annealing and exciton diffusion, we have gained a deeper understanding of how quantum processes might facilitate efficient energy transfer and information processing. The exploration of multiple pathways simultaneously, whether in light-harvesting complexes or quantum annealers, highlights the robustness and optimization capabilities inherent in quantum systems.

Looking ahead, further research could focus on quantifying the impact of noise on quantum annealing systems and exploring strategies to mitigate decoherence at higher temperatures. Additionally, experimental validation of the theoretical models presented here, particularly through implementation on quantum hardware like D-Wave systems, could provide valuable insights into the practical efficacy of quantum annealing for solving complex optimization problems. On the light-harvesting systems side, efforts are underway to deliberately design—and evaluate via simulations—the nano-structures of organic solar cells so as to maximize transfer efficiency [63, 64, 65]. Other research efforts explore the role the delocalized states and coherent dynamics

can potentially play in energy storage [66] and in the acceleration of chemical reactions [67] and nuclear reactions [68].

As we stand at the intersection of quantum information science and energy transfer dynamics, the potential for cross-disciplinary advancements is immense. By leveraging some of the principles elucidated in this study, research can continue to push the boundaries of quantum technologies, paving the way for innovative solutions in optimization, energy harvesting, energy conversion, and quantum computation. The journey towards harnessing the power of delocalized states in quantum systems is just beginning, promising exciting developments and transformative applications in the realms of quantum information and energy science.

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